7 A student sets up the circuit shown in Fig. 7.1.

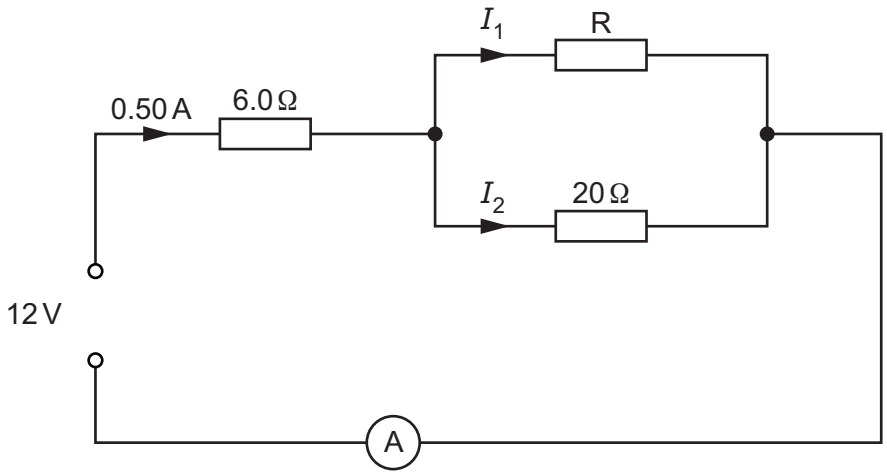

Fig. 7.1

(a) Determine the value of the current measured by the ammeter.

current = ... [1]

(b) Calculate the potential difference (p.d.) across the 6.0 Ω resistor.

p.d. = ... [2]

(c) Show that the p.d. across the 20 Ω resistor is 9.0 V.

[1]

(d) Calculate the resistance of resistor R.

resistance = ... [3]

[Total: 7]

DO NOT WRITE IN THIS MARGIN